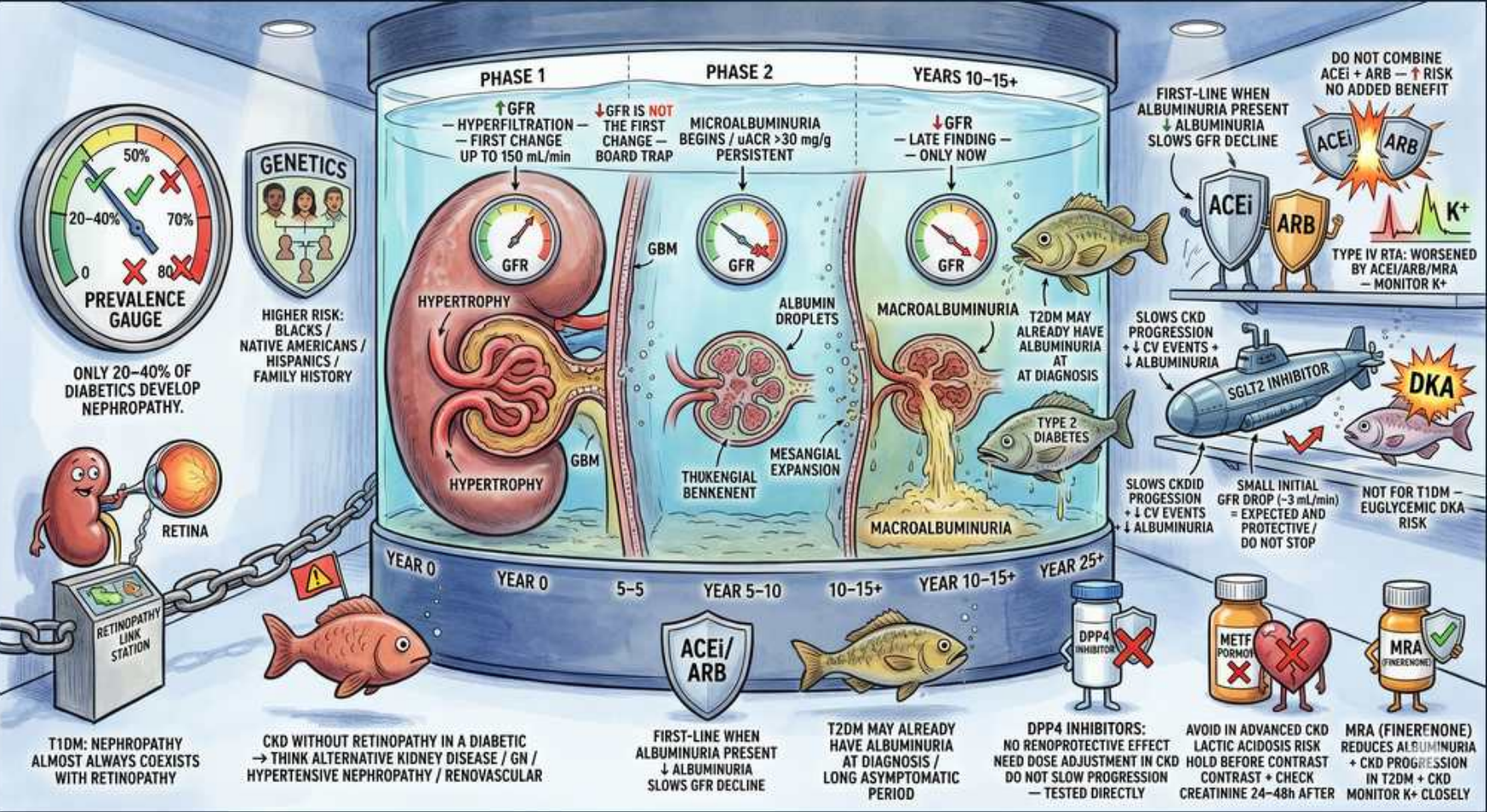


Diabetes — Complication 1: Diabetic Nephropathy



1. Prevalence gauge

WHAT YOU SEE

Gauge: only 20-40% of diabetics develop nephropathy.

WHAT IT ENCODES

Only ~20-40% of diabetics develop diabetic kidney disease — it is not universal, but diabetic nephropathy is the leading cause of ESRD/dialysis in the US. Higher risk in Black, Hispanic, and Native American patients and with poor glycemic/BP control and family history. Screen annually with spot urine ACR and serum creatinine/eGFR.

BOARD TRAP

WRONG: 'Every diabetic inevitably progresses to dialysis, so screening changes nothing.' Only a subset progress, and screening plus ACEi/ARB, SGLT2i, glycemic and BP control sharply slow progression. Discriminator: nephropathy is preventable/modifiable, and annual ACR + eGFR screening starts at diagnosis in T2DM and 5 years after diagnosis in T1DM.

2. Phase 1 — hyperfiltration

WHAT YOU SEE

Phase 1: GFR INCREASED (hyperfiltration), first change, up to 50% increase.

WHAT IT ENCODES

The EARLIEST change is glomerular HYPERFILTRATION: GFR rises (up to ~50% above normal) from afferent arteriolar vasodilation and efferent constriction (angiotensin II), raising intraglomerular pressure. This is accompanied by renal/glomerular hypertrophy. Chronic hyperfiltration drives later structural damage.

BOARD TRAP

WRONG: 'The first sign of diabetic nephropathy is a falling GFR.' Early GFR is HIGH (hyperfiltration); GFR decline is a LATE event. Discriminator: hyperfiltration (increased GFR) is the first change; a low eGFR early on points away from early diabetic nephropathy.

3. Hypertrophy / mesangial expansion

WHAT YOU SEE

Hypertrophy, GBM thickening, mesangial expansion, Kimmelstiel-Wilson nodules.

WHAT IT ENCODES

Structural lesions: glomerular basement membrane (GBM) thickening, mesangial matrix expansion, and Kimmelstiel-Wilson nodular glomerulosclerosis (PAS-positive eosinophilic nodules) — the pathognomonic finding. Hyaline arteriosclerosis of BOTH afferent and efferent arterioles is characteristic of diabetes.

BOARD TRAP

WRONG: 'Kimmelstiel-Wilson nodules are seen in any cause of nephrotic syndrome.' Nodular glomerulosclerosis with K-W nodules is specific to diabetic nephropathy (mimicked only by amyloid/MPGN/light-chain disease). Discriminator: K-W nodules + GBM thickening + mesangial expansion = diabetic nephropathy.

4. Phase 2 — microalbuminuria

WHAT YOU SEE

Phase 2: microalbuminuria begins, ACR 30-300 mg/g, persistent.

WHAT IT ENCODES

First detectable LAB marker = moderately increased albuminuria ('microalbuminuria'), urine albumin-to-creatinine ratio (ACR) 30-300 mg/g, typically appearing 5-10 years after onset. Must be PERSISTENT (2 of 3 samples over 3-6 months positive), excluding transient causes (exercise, fever, UTI, CHF, hyperglycemia, menstruation).

BOARD TRAP

WRONG: 'A single elevated ACR of 120 mg/g confirms diabetic nephropathy.' One positive sample is insufficient and can be transient; confirm with 2 of 3 over 3-6 months. Discriminator: diagnosis of persistent albuminuria requires repeat confirmation, excluding reversible causes.

5. Macroalbuminuria — late

WHAT YOU SEE

Phase years 10-15+: GFR decline, macroalbuminuria, late finding.

WHAT IT ENCODES

Overt nephropathy = severely increased albuminuria/macroalbuminuria (ACR >300 mg/g, may reach nephrotic-range >3.5 g/day) with progressive GFR decline, usually 10-15+ years in. Often accompanied by worsening hypertension and edema. Albuminuria precedes and predicts the GFR fall.

BOARD TRAP

WRONG: 'GFR decline appears before any albuminuria in diabetic nephropathy.' Albuminuria (micro then macro) precedes the GFR decline in the classic course. Discriminator: the sequence is hyperfiltration -> microalbuminuria -> macroalbuminuria -> falling GFR; declining GFR is the late event.

6. ACEi / ARB first-line

WHAT YOU SEE

ACEi/ARB: first-line when albuminuria present, slows GFR decline.

WHAT IT ENCODES

ACE inhibitor or ARB is FIRST-LINE once albuminuria is present (and the antihypertensive of choice in diabetics with albuminuria): they reduce intraglomerular pressure via efferent arteriolar dilation, lowering albuminuria and slowing GFR decline independent of BP. Target BP generally <130/80.

BOARD TRAP

WRONG: 'Combine an ACEi AND an ARB for additive renoprotection in heavy albuminuria.' Dual RAAS blockade increases hyperkalemia and AKI without benefit (ONTARGET/VA NEPHRON-D) and is contraindicated. Discriminator: use ONE RAAS agent (ACEi OR ARB), never both together.

7. Type II RTA (proximal)

WHAT YOU SEE

Type II RTA worsened by ACEi/ARB; monitor K⁺ (hyperkalemia from type IV RTA).

WHAT IT ENCODES

Diabetes classically causes TYPE IV RTA (hyporeninemic hypoaldosteronism) -> hyperkalemia with mild non-anion-gap metabolic acidosis. ACEi/ARB (and MRAs) further reduce aldosterone and raise potassium, so check potassium and creatinine 1-2 weeks after starting or up-titrating.

BOARD TRAP

WRONG: 'Hyperkalemia in a diabetic on an ACEi reflects type I (distal) RTA.' Diabetic hyperkalemic RTA is type IV (hyporeninemic hypoaldosteronism); type I distal RTA causes HYPOkalemia. Discriminator: diabetes + hyperkalemia + mild NAGMA = type IV RTA; monitor K⁺ on RAAS blockade.

8. SGLT2i renoprotection

WHAT YOU SEE

SGLT2i submarine: slows CKD progression, reduces albuminuria.

WHAT IT ENCODES

Add an SGLT2 inhibitor for renoprotection on top of a maximally tolerated ACEi/ARB: by restoring tubuloglomerular feedback (afferent vasoconstriction) it lowers intraglomerular pressure, reduces albuminuria, and slows CKD progression and CV/HF events (CREDENCE, DAPA-CKD) — benefit extends even to non-diabetic CKD.

BOARD TRAP

WRONG: 'SGLT2 inhibitors should be started in a T1DM patient for kidney protection.' SGLT2i are not approved in T1DM due to euglycemic DKA risk. Discriminator: SGLT2i renoprotection applies to T2DM/CKD; in T1DM the euglycemic DKA risk outweighs benefit.

9. Small initial eGFR dip

WHAT YOU SEE

Small initial eGFR drop (~3 mL/min) expected and protective; do not stop drug.

WHAT IT ENCODES

A modest initial eGFR dip (or creatinine rise up to ~30%) after starting an ACEi/ARB or SGLT2i is EXPECTED and reflects reduced intraglomerular pressure — it is hemodynamic and ultimately renoprotective. Recheck in 1-2 weeks; the eGFR stabilizes and the long-term slope improves.

BOARD TRAP

WRONG: 'Stop the ACEi immediately because creatinine rose 20% after starting it.' A rise up to 30% is expected and the drug should be continued. Discriminator: only investigate/stop (think bilateral renal artery stenosis, volume depletion) if creatinine rises >30% or potassium becomes dangerous.

10. Retinopathy coexists

WHAT YOU SEE

Retina image: T1DM nephropathy almost always coexists with retinopathy.

WHAT IT ENCODES

In T1DM, diabetic nephropathy almost ALWAYS coexists with retinopathy (shared microvascular pathology); the absence of retinopathy in a T1DM patient with renal disease strongly suggests a non-diabetic/alternative cause. The link is looser in T2DM but retinopathy still supports a diabetic etiology.

BOARD TRAP

WRONG: 'A T1DM patient with proteinuria, hematuria, and NO retinopathy has diabetic nephropathy.' Lack of retinopathy (plus active sediment/hematuria) points to non-diabetic kidney disease and warrants biopsy. Discriminator: diabetic nephropathy in T1DM tracks with retinopathy; its absence -> look for alternative renal disease.

11. DPP-4 / dose adjust in CKD

WHAT YOU SEE

DPP-4 no renoprotective effect, needs renal dose adjustment in CKD; long asymptomatic period.

WHAT IT ENCODES

DPP-4 inhibitors lower glucose but have NO independent renoprotective effect; most require renal dose reduction in CKD (sitagliptin, saxagliptin, alogliptin) — EXCEPT linagliptin, which is hepatically eliminated and needs no renal adjustment. They are weight-neutral with low hypoglycemia risk.

BOARD TRAP

WRONG: 'Use a DPP-4 inhibitor to slow diabetic CKD progression.' DPP-4i do not protect the kidney; only ACEi/ARB, SGLT2i, GLP-1 RA, and finerenone do. Discriminator: for renoprotection choose SGLT2i/GLP-1 RA/MRA; DPP-4i is purely glycemic (and use linagliptin if no renal dose adjustment is desired).

12. Avoid metformin advanced CKD

WHAT YOU SEE

Metformin: avoid advanced CKD (lactic acidosis); adjust eGFR 30-45; stop <30; check Cr 24-48h after contrast.

WHAT IT ENCODES

Metformin risks lactic acidosis when it accumulates in renal failure: do not START if eGFR <45, REDUCE dose for eGFR 30-45, and STOP if eGFR <30. Hold before/at the time of iodinated contrast if eGFR <30 (or for any AKI risk) and recheck creatinine 48 h after contrast before resuming.

BOARD TRAP

WRONG: 'Continue full-dose metformin through a CT with IV contrast in a patient with eGFR 28.' Metformin must be held around contrast when eGFR <30 (AKI -> lactic acidosis) and resumed only after renal function is confirmed stable. Discriminator: stop metformin if eGFR <30 and around contrast; recheck Cr at 48 h.

13. MRA (finerenone)

WHAT YOU SEE

MRA / finerenone: reduces albuminuria + CKD progression; monitor K+ closely.

WHAT IT ENCODES

Finerenone, a NONSTEROIDAL selective mineralocorticoid-receptor antagonist, reduces albuminuria, slows CKD progression, and lowers CV events in T2DM with CKD (FIDELIO/FIGARO-DKD) on top of ACEi/ARB +/- SGLT2i. It is added for residual albuminuria; check baseline potassium and eGFR before starting.

BOARD TRAP

WRONG: 'Start finerenone in a diabetic CKD patient with serum potassium 5.6 mEq/L to add renoprotection.' Finerenone causes hyperkalemia and should not be started when potassium is already elevated (>5.0). Discriminator: confirm K+ <=5.0 and acceptable eGFR before starting finerenone, and monitor potassium closely, especially with concurrent ACEi/ARB.